CPT107

Part I. Number Systems and Proof Techniques

1. Natural Numbers N = {0, 1, 2, 3,...}

2. Integers Z (Positive Integers Z+)

3. Rational Numbers Q (can be written as x/y where x and y are integers and y is not 0)

4. Real Numbers R (rational or irrational)

5. Prime Numbers (greater than 1, has exactly two positive integer divisors: 1 and itself)

   Every integer greater than 1 can be written as a unique product of prime numbers.

   A positive integer is even if it has 2 as a factor. Otherwise, it is odd.

6. Find a counter-example

7. Proof by contradiction

8. Proof by induction

Base Case: Show that the property holds for $n = 0$.

Inductive Step: Assume that the property holds for $n = m$. Show that it holds for $n = m + 1$.

Conclusion: You can now conclude that the property holds for every natural number n.

Part II. Set Theory

1. Equality {1} = {1, 1, 1} {5, 4, 4, 3, 5} = {3, 4, 5}

2. Union A or B

3. Intersection A and B

4. Relative Complement A – B

5. Complement ~A

6. Symmetric Difference A△B [(A∪B) – (A∩B)]

7. Commutative Laws(交换律):

   A∪B = B∪A, A∩B = B∩A

   Associative Laws(结合律):

   A∪(B∪C) = (A∪B)∪C, A∩(B∩C) = (A∩B)∩C

Identity Laws(恒等律):

$A \cup \emptyset = A$, $A \cup U = U$, $A \cap \emptyset = \emptyset$, $A \cap U = A$

Distributive Laws(分配律):

$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$, $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Complement Laws(补集律):

$A \cup \sim A = U$, $A \cap \sim A = \emptyset$, $\sim U = \emptyset$, $\sim\emptyset = U$, $\sim(\sim A) = A$

De Morgan's Laws:

$\sim(A \cup B) = \sim A \cap \sim B$, $\sim(A \cap B) = \sim A \cup \sim B$

8. Power Set (the set of all subsets)

$Pow(A \cap B) = Pow(A) \cap Pow(B)$

$Pow(A \cup B) \neq Pow(A) \cup Pow(B)$

9. Cardinality (the number of elements)

$|A \cup B| = |A| + |B| - |A \cap B|$

$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$

10. Cartesian Product $A \times B$

11. Bit Strings

Bit strings can be used to represent the subsets of a set.

Suppose we have a set $S = \{s_1, \ldots, s_n\}$.

(I have given the n elements of the set names. s_1 is the name of the first element. s_2 is the name of the second element (which is different from the first element) and so on.)

If we have a subset $A \subseteq S$, the characteristic vector of A is the n-bit string $(b_1, \ldots, b_n) \in B^n$ where

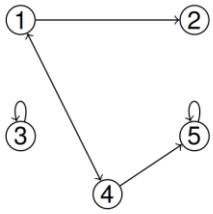
$$b_i = \begin{cases} 1 & \text{if} \quad s_i \in A \\ 0 & \text{if} \quad s_i \notin A \end{cases}$$

Part III. Relations

1. Directed Graph(or Digraph)

Consider the relation $V \subseteq A \times A$ where $A = \{1, 2, 3, 4, 5\}$ and

$$V = \{(1,2), (3,3), (5,5), (1,4), (4,1), (4,5)\}.$$



2. Matrix

- We represent R by an array M of n rows and m columns. Such an array is called a *n by m matrix*.
- The entry in row i and column j of this matrix is given by $M(i, j)$ where

$$M(i,j) = \begin{cases} T & \text{if} \quad (a_i, b_j) \in R \\ F & \text{if} \quad (a_i, b_j) \notin R \end{cases}$$

Let $A = \{1, 3, 5, 7\}$, $B = \{2, 4, 6\}$, and

$$U = \{(x, y) \in A \times B \mid x + y = 9\}$$

Assume an enumeration $a_1 = 1$, $a_2 = 3$, $a_3 = 5$, $a_4 = 7$ and $b_1 = 2$, $b_2 = 4$, $b_3 = 6$. Then M represents U, where

$$M = \begin{bmatrix} F & F & F \\ F & F & T \\ F & T & F \\ T & F & F \end{bmatrix}$$

3. Infix Notation xRy (x, y) (x is R-related to y)

4. Reflexive, Symmetric, Antisymmetric, Transitive

A binary relation R on a set A is

- *reflexive* when xRx for all $x \in A$.
- *symmetric* when xRy implies yRx for all $x, y \in A$;
- *antisymmetric* when xRy and yRx imply $x = y$ for all $x, y \in A$;
- *transitive* when xRy and yRz imply xRz for all $x, y, z \in A$.

In the directed graph representation, R is

- *reflexive* if there is always an arrow from every vertex to itself;
- *symmetric* if whenever there is an arrow from x to y there is also an arrow from y to x;
- *antisymmetric* if whenever there is an arrow from x to y and $x \neq y$, then there is no arrow from y to x;
- *transitive* if whenever there is an arrow from x to y and from y to z there is also an arrow from x to z.

5. Closure (reflexive, symmetric or transitive)

6. Equivalence Relation (reflexive, symmetric and transitive)

7. Equivalence Class

Define a relation R on the set $\mathbb{R}$ of real numbers by setting xRy if and only if $x - y$ is an integer. Prove that R is an equivalence relation. Moreover,

- $E_0 = \mathbb{Z}$ is the equivalence class of 0;
- $E_{\frac{1}{2}} = \{\ldots, -2\frac{1}{2} - 1\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, 1\frac{1}{2}, 2\frac{1}{2}, \ldots\}$ is the equivalence class of $\frac{1}{2}$.

8. Partial Ordering(or Partial Order) (reflexive, antisymmetric and transitive)

9. Partially Ordered Set(or Poset) $(S, \succ)$

10. Comparable or Incomparable

- The elements a and b of a poset $(S, \succ)$ are called ***comparable*** if either $a \succ b$ or $b \succ a$.

- When a and b are elements of S such that neither $a \succ b$ nor $b \succ a$, a and b are called ***incomparable***.

11. Predecessor, Immediate Predecessor and Successor

- Let $(\mathbf{Z}, \leq)$ be a partially ordered set. If $x \leq y$ but $x \neq y$, we write $x < y$ and say that x is a **predecessor** of y or y is a **successor** of x.

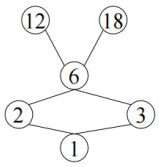
- A given y may have many predecessors, but if $x < y$ and there is no z with $x < z < y$, then x is an **immediate predecessor** of y.

12. Hasse Diagram (visual depiction of a partially ordered set)

Two nodes in a Hasse diagram should never be joined by a horizontal line.

Draw the Hasse diagram for the relation “x divides y” on {1,2,3,6,12,18}.

1. Write the ordered pairs (x,y) of this relation.
2. Find the predecessors.
3. Find the immediate predecessors.
4. Draw the diagram.



13. Totally Ordered(or Linearly Ordered)

- If $(S,\succ)$ is a poset and every two elements of S are *comparable*, S is called totally *ordered* or *linearly ordered* set, and $\succ$ is called a ***total order*** or a ***linear order***.
- A totally ordered set is also called a chain.

14. Inverse Relation

15. Composition of Relations

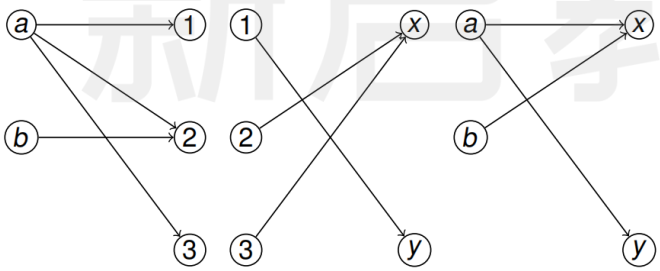
Then the entries $P(i,j)$ of the logical matrix P representing $S \circ R$ are given by

$P(i,j) = T$ if there exists l with $1 \le l \le m$ such that $M(i,l) = T$ and $N(l,j) = T$.

$P(i,j) = F$, otherwise.

We write $P = MN$.

Now let’s go back and see how this works for matrices representing relations



$$R : \begin{bmatrix} T & T & T \\ F & T & F \end{bmatrix} \quad S : \begin{bmatrix} F & T \\ T & F \\ T & F \end{bmatrix} \quad S \circ R : \begin{bmatrix} T & T \\ T & F \end{bmatrix}$$

Part IV. Functions and PHP

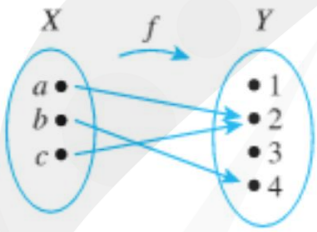
1. Function (every element of A is associated with a uniquely specified element of B)

2. Domain, Codomain, Range

Suppose $f: A \to B$.

- A is called the ***domain*** of f.
- B is called the ***codomain*** of f.
- The ***image of x under f*** is the value $f(x)$;
- if R is any subset of A, then $f(R) = \{f(r) \mid r \in R\}$ is **the set of images of elements of *R*** and is called **the *image of R***.
- The ***image*** *or* ***range f(A) of f*** is $f(A) = \{f(x) \mid x \in A\}$

Example



domain of $f = \{a, b, c\}$,

co-domain of $f = \{1, 2, 3, 4\}$

range of $f = \{2, 4\}$

3. Injective(one-to-one)

Definition. Let $f: A \to B$ be a function. We call f an ***injective*** (or **one-to-one**) function if

$$f(a_1) = f(a_2) \Rightarrow a_1 = a_2 \text{ for all } a_1, a_2 \in A.$$

If the function f: $\mathbf{R} \to \mathbf{R}$ is defined by the rule $f(x) = 4x - 1$, for each real number x, then f is one-to-one.

Proof: Suppose x_1 and x_2 are real numbers such that $f(x_1) = f(x_2)$. *[We must show that $x_1 = x_2$.]* By definition of f,

$$4x_1 - 1 = 4x_2 - 1.$$

Adding 1 to both sides gives

$$4x_1 = 4x_2,$$

and dividing both sides by 4 gives

$$x_1 = x_2,$$

[as was to be shown].

4. Surjective(onto)

Definition. f : A → B is **surjective** (or onto) if the range of *f* coincides with the codomain of *f*.

This means that for every b ∈ B there exists a ∈ A with b = f(a).

If $f: \mathbf{R} \to \mathbf{R}$ is the function defined by the rule $f(x) = 4x - 1$ for each real number x, then f is onto.

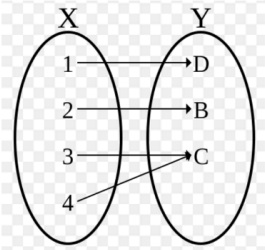
Proof: Let $y \in \mathbf{R}$. *[We must show that $\exists x$ in $\mathbf{R}$ such that $f(x) = y$.]* Let $x = (y+1)/4$. Then x is a real number since sums and quotients (other than by 0) of real numbers are real numbers. It follows that

$$\begin{aligned} f(x) &= f\left(\frac{y+1}{4}\right) && \text{by substitution} \\ &= 4\cdot\left(\frac{y+1}{4}\right) - 1 && \text{by definition of } f \\ &= (y+1) - 1 = y && \text{by basic algebra,} \end{aligned}$$

[as was to be shown].

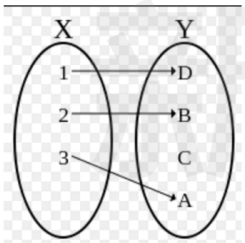
5. Bijective (we call f bijective if f is both injective and surjective)

满射：对任意b，存在a满足f(a) = b～



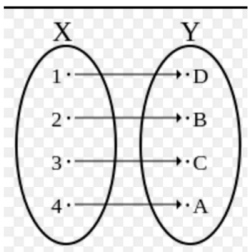
即：值域y是满的，每个y都有x对应，不存在某个y没有x对应的情况～

单射：(one-to-one function) 一对一函数，x不同则y不同～



即：没有一个x对应两个y，也没有一个y有对应两个x～

双射：既是满射，也是单射～



即：每个y都有x对应，而且都是一一对应～

6. Inverse Functions (a function f is invertible if and only if it is a bijection)

7. Composition g o f

8. The Pigeonhole Principle

Consider a function $f : A \to B$ where A and B are finite sets and
$|A| > k|B|$ for some natural number k.
Then, there is a value of f which occurs at least $k + 1$ times.

Part V. Proposition Logic and Predicate Logic

1. Truth Values 0-false 1-true Truth Table

2. Negation ¬P

3. Conjunction P∧Q (1 if P = 1 and Q = 1)

4. Disjunction P∨Q (1 if P = 1 or Q = 1)

5. P→Q = ¬P∨Q (if P is true, then Q is true)

P→Q is true always except when P is true and Q is false.

P	Q	$\neg P$	$(\neg P \vee Q)$
1	1	0	1
1	0	0	0
0	1	1	1
0	0	1	1

6. P↔Q = ((P→Q)∧(Q→P)) (P is true if and only if Q is true)

P↔Q is true when P and Q have the same truth value.

P	Q	$(P \to Q)$	$(Q \to P)$	$((P \to Q) \wedge (Q \to P))$
1	1	1	1	1
1	0	0	1	0
0	1	1	0	0
0	0	1	1	1

7. Tautology (is true under all interpretations)

⊤ = (p∨¬P) I(p∨¬P) = 1

8. Contradiction (is false under all interpretations)

⊥ = (p∧¬P) I(p∧¬P) = 1

9. Semantic Consequence

Definition Suppose Γ is a finite set of formulas and P is a formula. Then P follows from Γ ("is a semantic consequence of Γ") if the following implication holds for every interpretation I:

$$\text{If } I(Q) = 1 \text{ for all } Q \in \Gamma, \text{ then } I(P) = 1.$$

This is denoted by

$$\Gamma \models P.$$

10. Logical Equivalence

Definition Two formulas P and Q are called equivalent if they have the same truth value under every possible interpretation. In other words, P and Q are equivalent if $I(P) = I(Q)$ for every interpretation I. This is denoted by

$$P \equiv Q.$$

11. Commutative Laws(交换律):

$(P \vee Q) \equiv (Q \vee P)$, $(P \wedge Q) \equiv (Q \wedge P)$

Associative Laws(结合律):

$(P \vee (Q \vee R)) \equiv ((P \vee Q) \vee R)$, $(P \wedge (Q \wedge R)) \equiv ((P \wedge Q) \wedge R)$

Identity Laws(恒等律):

$(P \vee \top) \equiv \top$, $(P \vee \bot) \equiv P$, $(P \wedge \top) \equiv P$, $(P \wedge \bot) \equiv \bot$

Distributive Laws(分配律):

$P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$, $P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$

Complement Laws(补集律):

$P \vee \neg P \equiv \top$, $P \wedge \neg P \equiv \bot$, $\neg\bot \equiv \top$, $\neg\top \equiv \bot$, $\neg\neg P \equiv P$

De Morgan's Laws:

$\neg(P \vee Q) \equiv (\neg P \wedge \neg Q)$, $\neg(P \wedge Q) \equiv (\neg P \vee \neg Q)$

12. The relation $\equiv$ is an equivalence relation.

13. Signatures

Signatures

Definition

A signature S is a set consisting of:

- predicate symbols, each with an associated arity
- function symbols, each with an associated arity
- constant symbols

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Syntax of predicate logic
Signatures

- A signature defines the "vocabulary" relative to which formulae can be expressed.
- It is the set of all the symbols that can be used in a formula:
  - Predicate symbols: *Student*, *Younger*, *Bird*, …
  - Constant symbols: *alice*, *comp118*, *42*, …
  - Function symbols: *Mother*, *Plus*, *Minus*, …
- It also fixes an arity (number of parameters) for each predicate and function symbol in it.

  Example: In the formulae of the last lecture, we assumed that *Student* and *Younger* have arity 1 and 2, respectively.

14. Proposition Logic vs. Predicate Logic

Proposition logic vs *Predicate logic*

- in **Proposition logic**, we can express statements as a whole, and combinations of them. A proposition is a statement that is having a *truth value* (either true or false) associated with it.

- in **Predicate Logic** (FOL) a statement has a specific inner structure, consisting of terms and predicates.

Terms (variable and functions) denote objects in some reality, and *predicates* express properties of, or relations between those objects.

The truth value is dependent upon the variables.

Proposition logic vs *Predicate logic*

- A predicate is a sentence that contains one or more variables
- A predicate is neither true nor false
- A predicate becomes a proposition when the variables are substituted with specific values
- The domain of a predicate variable is the **set of all values that may be substituted for the variable**

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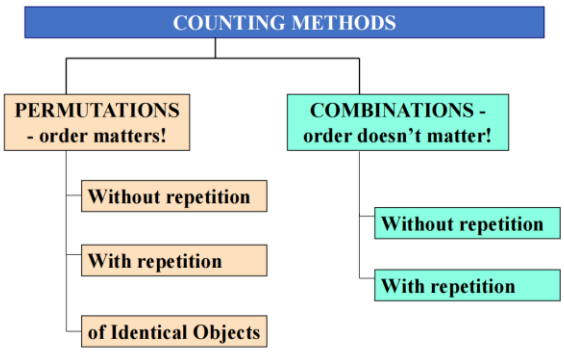
Part VI. Combinatorics and Probability

1. Basic Counting Principles and Counting Rules

   Product Rule(Multiplication Principle), Sum Rule(Addition Principle),

   Subtraction Rule(Subtraction Principle), Division Rule(Division Principle)

2. Counting Methods



3. Permutations (Order matters!)

Permutations - without repetition

This means that *once one of the elements of the set has left, it cannot reappear.*

Formula for **permutations without repetition**

$$P(n,r) = \frac{n!}{(n-r)!}$$

Permutation with repetition

A permutation with repetition is the one in which we have n elements that we take a number of r times.

Permutation with repetition, *if we pick up an element, it can come back out again in the second election and so on until reaching r times.*

Formula for **permutations with repetition**

$$P(n,r) = n^r$$

Permutations of Identical Objects

The number of permutations of n objects with n_1 identical objects of type 1, n_2 identical objects of type 2, …, and n_k identical objects of type k is:

$$\frac{n!}{n_1!\,n_2!\,\ldots n_k!}$$

In general, the number of permutations of n objects with r of the objects identical objects is:

$$\frac{n!}{r!}$$

4. Combinations

Combinations – without repetition

- An unordered selection of r elements from a set containing n distinct elements.

Example:

S = {A, J, Q, K}. {A, J, K}={K, A, J}={J, K, A} are all *3-combinations of set S*.

The number of r-combinations of a set of size n is called **binomial coefficient**, denoted $C(n,r)$, and given by:

$$C(n,r) = \binom{n}{r} = \frac{n!}{r!\,(n-r)!}$$

$0 \le r \le n$

Combinations – with repetition

- An **unordered** selection of r elements from a set containing n distinct elements, with **repetition permitted**.

$$C(n+r-1,r) = \binom{n+r-1}{r} = \frac{(n+r-1)!}{r!\,(n-1)!}$$

5. Mutually Exclusive Events or Mutually Inclusive Events

6. Conditional Probability

Conditional Probability

The probability of occurrence of any event X when another event Y in relation to X has already occurred is known as conditional probability.

If the probability of "**X**" is conditional (affected by) "**Y**" (what happens before it) then we say:

The probability of "X" given "Y"

$$P(X \mid Y)$$

The probability that this **WILL** happen

Assuming this already **HAS** happened

$$P(X \mid Y) = \frac{\frac{N(X \cap Y)}{N}}{\frac{N(Y)}{N}} = \frac{P(X \cap Y)}{P(Y)}$$